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Evaluating a Virtual Knowledge Graph for Federating Research Information Systems at the Institute of Technology of Cambodia

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Abstract: *Research information at universities is commonly distributed across independent operational systems, which makes integrated discovery of researchers, academic records, and research activities highly challenging. This study evaluates a Virtual Knowledge Graph (VKG) approach for integrating research-management data at the Institute of Technology of Cambodia (ITC) without replicating source databases into a separate Resource Description Framework (RDF) store. The proposed architecture federates relational PostgreSQL-based institutional systems via Trino, and exposes the data via a unified semantic layer using Ontology-Based Data Access (OBDA) mappings and a SPARQL endpoint with Ontop. A domain ontology represents core institutional entities and their relationships, including researchers, departments, academic records, and research projects. The approach was evaluated using three representative query groups: single-source SMIS queries, SMIS-Portal federated queries, and SMIS-RMIS/RIC research-information queries. Each configuration was measured across 1 to 7 verified Trino workers using 40 randomised runs after warm-up on an Ubuntu Server virtual machine with direct source-database connectivity. All 840 measured requests completed successfully with stable result counts. The worker count associated with the lowest observed mean depended on the workload: one worker produced the lowest observed mean for the SMIS-only and SMIS-RMIS/RIC workloads, while three workers produced the lowest observed mean for the SMIS-Portal workload. No formal hypothesis test was performed; therefore, the observed differences are not claimed as statistically significant improvements. This study demonstrates that an OBDA-based VKG can provide direct, non-materialised semantic access to institutional research data while preserving existing source systems. The architecture is cost-conscious because it avoids the replication, synchronisation, and maintenance of a separate materialised RDF store. Future work will expand this framework toward external identity-linking validation.*

Keywords: Ontology-Based Data Access; Ontop; Research Information Systems; Trino; Virtual Knowledge Graph

1. INTRODUCTION

Research information in higher education institutions is commonly distributed across independent operational systems. Researcher profiles, student and employee records, departments, projects, and publication information are often maintained in separate databases for different administrative purposes. This fragmentation makes it difficult to obtain an integrated view of institutional research activities, identify collaboration opportunities, and prepare reliable reports.

At the Institute of Technology of Cambodia (ITC), relevant information is distributed across the School Management Information System (SMIS), the Student Portal, and the Research Management Information System managed by the Research and Innovation Center (RMIS/RIC). These systems were developed independently and do not share a common semantic model or canonical identifiers for all entities. Consequently, answering questions that span multiple systems requires manual cross-referencing, custom database queries, or physical data replication.

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Ontology-Based Data Access (OBDA) addresses this problem by exposing relational data through a conceptual ontology without requiring complete replication into a separate Resource Description Framework (RDF) store [4,5]. RDF provides a standard graph-based model for representing identified resources and their relationships [7]. Furthermore, RDF allows one to build a decentralised web of knowledge; using global identifiers, one can create links with other datasets to enrich the data. RDF enables links to existing resources without physically integrating them. These resources and their relationships can be queried using SPARQL. In a Virtual Knowledge Graph (VKG), SPARQL queries are translated through declarative mappings into queries over the original relational data sources [5,10,15]. This approach can preserve existing institutional databases while providing a unified semantic access layer. By querying data in place, the approach also avoids operating an additional synchronised RDF repository. This is a cost-conscious architectural benefit rather than a measured financial-cost comparison.

This study evaluates a VKG architecture for ITC research information systems. Trino[2] is used as the distributed SQL federation layer for the institutional PostgreSQL sources. It supports cross-source SQL execution without requiring a centralised data warehouse [14]. Ontop exposes the mapped data through a SPARQL endpoint. The proposed architecture is designed for read-only semantic integration; it does not replace the existing operational systems or modify their source records.

This study evaluates whether a Trino- and Ontop-based Virtual Knowledge Graph can provide practical, direct, and non-materialised[3] semantic access to distributed ITC research-management data without copying source databases into a separate RDF store. The evaluation considers functional correctness, response time, and the effect of increasing the number of Trino workers. The study makes the following contributions:

1. It defines an institutional ontology and Ontop mapping layer for key research-information entities, including researchers, departments, academic records, research projects, and research articles.
2. It implements a non-replicating three-tier architecture based on PostgreSQL data sources, Trino federation, Ontop semantic virtualisation, Docker, and a SPARQL endpoint.
3. It evaluates three representative query workloads: SMIS-only queries, federated SMIS-Portal queries, and distributed SMIS-RMIS/RIC queries.
4. It evaluates the effect of 1 to 7 verified Trino workers on end-to-end response time for three representative workloads under a direct server-to-source deployment.

The scope of this study is the implementation and performance evaluation of direct, non-materialised semantic access to institutional data. It evaluates functional correctness, response time, and practical behaviour of the VKG architecture for selected read-only workloads. The goal of this study is to assess whether a VKG approach is a viable, extensible[4], and cost-conscious approach to conceptually integrating research-management data from multiple sources. It does not evaluate researcher identity matching, publication completeness, or data-quality correction across the underlying institutional systems. These topics require separately validated reference datasets and are outside the present scope of this study.

2. RELATED WORK

Ontology-Based Data Access (OBDA) provides a well-established approach for exposing relational data through an ontology and declarative mappings, without requiring complete replication into a separate RDF repository [4,5]. Virtual Knowledge Graph architectures apply this principle by translating SPARQL queries into queries over the original data sources.

Research-information integration commonly requires the combination of heterogeneous institutional systems and shared conceptual vocabularies. Established research-information models, such as CERIF and VIVO, support the representation and exchange of research-related entities and relationships [3,12]. However, adopting a common model alone does not solve the practical challenge of querying distributed operational databases without creating and maintaining additional copies.

This work applies these established OBDA and VKG principles to research-management integration at ITC. It constructs a Virtual Knowledge Graph over distributed PostgreSQL databases through a lightweight domain ontology and Ontop mappings [8]. Rather than extracting and materialising source records in a separate RDF repository, the approach rewrites SPARQL queries into federated SQL queries that are executed by Trino over the original SMIS, Student Portal, and RMIS/RIC databases [5,15].

The contribution is therefore an institutional implementation and end-to-end evaluation of this architecture

[2] Trino was previously named PrestoSQL and continues the Presto distributed-SQL lineage [14].

[3] "Direct and non-materialised access" is sometimes referred to as "live access." In this paper, "live" means that queries are evaluated against the current source systems.

[4] Here, in terms of adding and integrating additional databases.

for read-only semantic access. The evaluation verifies representative query results and measures response time across 1 to 7 Trino workers. It does not evaluate ontology alignment quality, author-identity matching, data completeness, concurrent-user throughput, or financial cost. These topics require additional reference datasets, workloads, or deployment evidence and are outside the scope of this study.

3. METHODOLOGY

This study adopts an Ontology-Based Data Access (OBDA) approach to integrate distributed research-information systems at the Institute of Technology of Cambodia (ITC). OBDA provides a semantic layer over relational databases through an ontology and declarative mappings, allowing users to query heterogeneous sources without physically copying the source records into a separate RDF repository [4,5]. The proposed method consists of four stages: source-system analysis, virtual knowledge graph construction, ontology mapping, and performance evaluation.

3.1 Virtual Knowledge Graph System Architecture

The proposed ITC Virtual Knowledge Graph (VKG) follows a three-tier architecture, as illustrated in **Fig. 1**. **Fig. 1** presents the system components and the separation between the operational data sources, the SQL federation layer, and the semantic access layer.

The first tier contains the operational PostgreSQL databases of three institutional systems: the School Management Information System (SMIS), the Student Portal, and the Research Management Information System operated by the Research and Innovation Center (RMIS/RIC). These systems remain independent and continue to serve their existing administrative functions. The proposed approach is non-intrusive: it does not modify the source databases. This is important, as the systems remain autonomous and independent from the VKG to be built.

The second tier uses Trino as a federated SQL layer. Trino connects to the institutional PostgreSQL sources through independent catalog configurations and presents them as a unified virtual relational environment. This federation layer supports cross-source queries without requiring a centralised data warehouse. The design follows the principles of distributed SQL query processing over multiple data sources [14].

The third tier uses Ontop to expose a semantic representation of the federated data. Ontop applies declarative mappings to translate SPARQL graph patterns into SQL queries that are executed through Trino over the original relational sources [5,15]. Client applications can therefore access integrated institutional data through a SPARQL

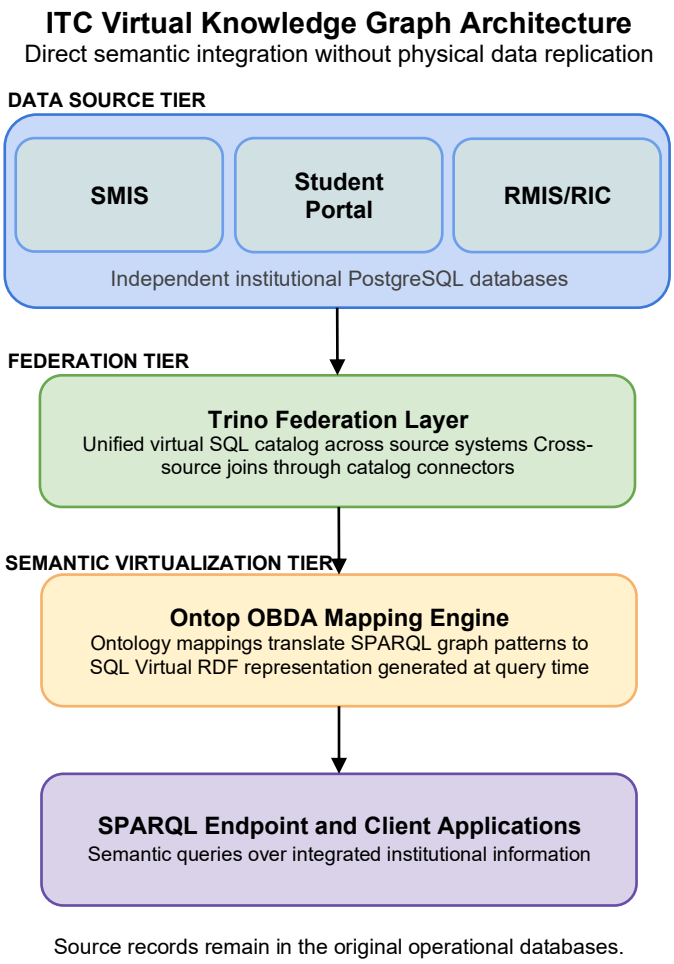


Fig. 1. Three-tier architecture of the ITC Virtual Knowledge Graph. Institutional PostgreSQL sources are federated by Trino, semantically mapped by Ontop, and exposed through a SPARQL endpoint.

endpoint while the underlying records remain in their original systems. The RDF data model and SPARQL query interface are W3C Recommendations [7,10], which are standards. **Fig. 2** illustrates the query-processing path. A client submits a SPARQL query to Ontop. Ontop uses the ontology and declarative mappings to rewrite the query into SQL and submits it to Trino. Trino executes the federated SQL over the required PostgreSQL sources and returns a relational result to Ontop. Ontop then serialises the result as the SPARQL query response.

3.2 Ontology and Mapping Design

A lightweight domain ontology was designed to represent the main entities needed for institutional research-information integration. The ontology includes the classes *Person*, *Employee*, *Student*, *Researcher*, *Department*,

Branch, *AcademicRecord*, *ResearchProject*, *ResearchProfile*, and *ResearchArticle*. The ontology currently uses the provisional namespace `http://smis.itc.edu.kh/ont#`. It is used consistently in the prototype mappings and queries, but it is not yet deployed as a dereferenceable Web resource. A persistent institutional namespace should be established before production deployment.

In a production deployment, ontology IRIs should resolve to documentation that describes the meaning of each class and property. This follows Linked Data principles, where IRIs can provide useful information about both the ontology and virtual data resources generated by Ontop [16]. These classes represent the shared conceptual vocabulary used across the source systems.

The main object properties include ***itc:hasDepartment***, ***itc:hasAcademicRecord***, ***itc:hasResearchIdentity***, and ***itc:participatesInProject***. Datatype properties are used for descriptive values such as names, identifiers, dates, titles, and organisational information. RDF resources are identified through stable IRIs, enabling records from different systems to be represented in a consistent graph model [7]. **Fig. 3** presents selected ontology classes and object properties used to create a common semantic view across the institutional systems.

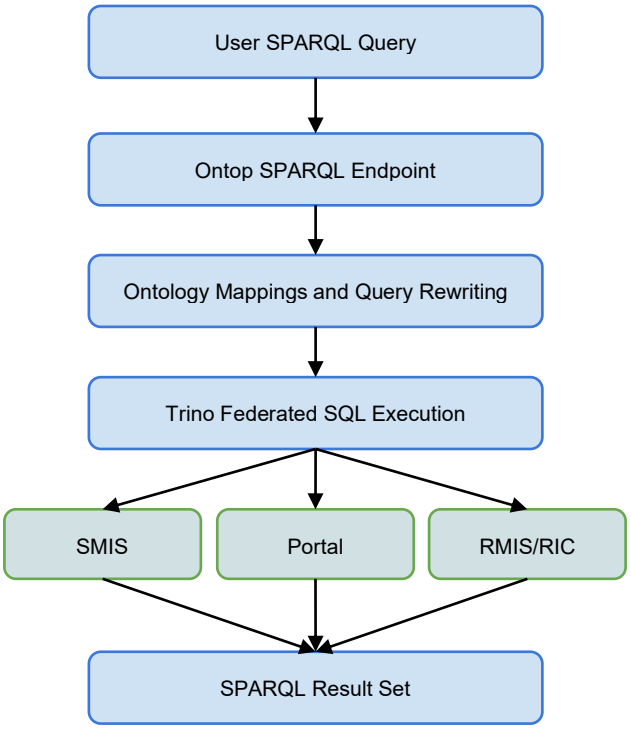


Fig. 2. Query-processing flow from a SPARQL request to federated SQL execution over the original PostgreSQL sources.

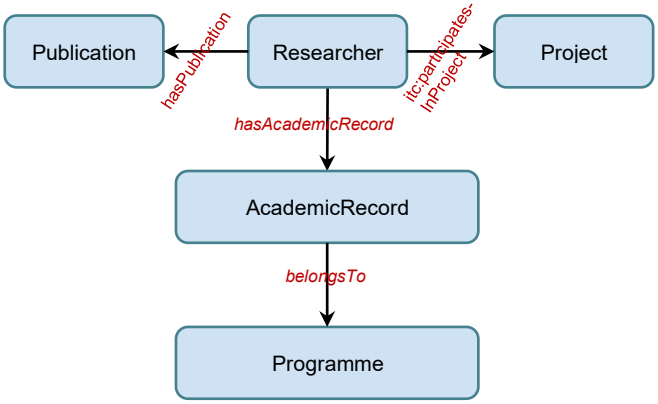


Fig. 3. Selected ontology classes and properties used in the ITC VKG prototype. The figure shows selected elements only; labels correspond to the ontology terms used in the mappings.

Ontop mappings connect ontology classes and properties to SQL queries over the federated Trino catalogs. Each mapping contains a source SQL query and a target RDF triple pattern. Entity IRIs are generated from stable source identifiers using consistent IRI templates. Where source identifiers use different native data types, the mappings explicitly cast identifier values to strings before IRI construction. This prevents inconsistent RDF identifiers for equivalent institutional entities. Although the mappings in this study use Ontop's native OBDA syntax, they serve the same source-to-target role as relational-to-RDF mapping approaches standardised by R2RML and Direct Mapping [6,8]. Earlier systems such as D2RQ also demonstrated virtual RDF access over relational databases [2].

The mapping layer is read-only. It does not modify SMIS, Student Portal, or RMIS/RIC records, and it does not materialise a separate RDF copy of the institutional data. Instead, RDF triples are generated virtually when a SPARQL query is executed. Therefore, relevant updates in the operational sources can be reflected through the semantic layer without an additional data-replication process.

3.3 Experimental Setup and Query Workloads

The performance benchmark consisted of repeated executions of the three SPARQL workloads defined in **Table 2**, using Trino configurations with 1 to 7 workers. It was executed on a dedicated Ubuntu Server 24.04 virtual machine with 16 virtual CPUs and 32 GB of provisioned memory, of which Linux reported 31 GiB as available host memory. The deployment used Docker and consisted of one Trino coordinator, one Ontop service, and between 1 and 7 Trino workers. The coordinator and each worker were limited to 3 GiB of container memory, while Ontop was limited to 2 GiB. The coordinator used one CPU identifier, each worker used two CPU identifiers, and one CPU identifier remained available for the operating system and Docker services. **Table**

1 summarises the server-VM environment and deployment configuration used in the experiment.

Table 1. Experimental environment and deployment configuration.

Component	Configuration
Host Operating System	Ubuntu Server 24.04
Deployment Platform	KVM/QEMU virtual machine; Docker Compose
Processor	16 virtual CPUs
Memory	32 GiB provisioned; 31 GiB reported by Linux
Trino Coordinator	3 GiB memory limit; one CPU identifier
Trino Worker	3 GiB memory limit; two CPU identifiers per worker (scaled 1 to 7)
Ontop Engine	2 GiB memory limit
Data Sources	Institutional PostgreSQL databases with direct connectivity
Worker Verification	Active Trino worker count checked before each benchmark block
Measured Repetitions	40 per workload and worker configuration (840 total)

The institutional PostgreSQL sources were accessed directly from the server VM. The experiment therefore evaluates end-to-end behaviour, including SPARQL request handling, Ontop query rewriting, Trino planning and scheduling, PostgreSQL access, cross-source joins, and result transfer. The experiment does not isolate the contribution of any individual component. Three query groups were designed to represent increasing integration complexity, as summarised in **Table 2**.

Table 2. SPARQL benchmark workloads used in the performance evaluation.

Integrated Sources	Purpose	Returned Rows
Q1: SMIS	Retrieve institutional researcher or staff information from one source.	9
Q2: SMIS and Student Portal	Join student or academic-record information across two systems.	20
Q3: SMIS and RMIS/RIC	Join researcher information with research-management data.	20

The workloads were designed to represent progressively more demanding access patterns. Q1 evaluates single-source semantic access. Q2 and Q3 evaluate federated access involving joins across institutional systems. The query result sets were intentionally kept stable through deterministic ordering so that worker configurations were evaluated using equivalent requests.

Three workloads were evaluated with 1 to 7 Trino workers. Each worker configuration received warm-up requests before measurement. For every workload and configuration, 40 measured repetitions were executed in a randomised order (i.e., all Q1, Q2, and Q3 queries were bundled together and shuffled) to reduce systematic effects from caching, temporary network variation, and remote database load. The primary performance experiment therefore comprises 840 planned SPARQL requests: 3 workloads x 7 worker configurations x 40 repetitions. Before performance measurement, each workload was functionally validated by comparing its returned bindings and row count with the expected source-system records.

3.4 Demonstration of Federated Semantic Access

To demonstrate the functional behaviour of the proposed VKG, **Listing 1** shows a shortened illustrative query based on workload Q2. It retains the same cross-source integration pattern as Q2 but applies LIMIT 5 solely to keep the example readable. It is not the benchmark query used to obtain the 20 returned rows reported for Q2 in **Table 2**. The query retrieves student resources, institutional identifiers, academic records, and associated academic information through the Ontop endpoint. Although the data remain in the original institutional sources, the client submits a single SPARQL query over the unified ontology vocabulary.

```
PREFIX itc: <http://smis.itc.edu.kh/ont#>
SELECT ?student ?idCard ?record ?academicYear
?department ?degree ?grade ?branch
WHERE {
 ?student a itc:Student ;
          itc:idCard ?idCard ;
          itc:hasAcademicRecord ?record .
          itc:forAcademicYear ?academicYear .
 OPTIONAL { ?record itc:hasDepartment
  ?department . }
 OPTIONAL { ?record itc:hasDegree
  ?degree . }
 OPTIONAL { ?record itc:hasGrade ?grade . }
 OPTIONAL { ?record itc:hasBranch ?branch .}
}
ORDER BY STR(?student) STR(?record)
LIMIT 5
```

Listing 1. Example federated SPARQL query for retrieving student and academic record information through the VKG endpoint.

The query result confirms that the VKG provides a single semantic access point for integrated information. To protect institutional confidentiality, **Table 3** presents selected synthetic bindings that illustrate the form of results returned by **Listing 1**; it does not reproduce source-system identifiers or values. This functional demonstration complements the performance evaluation reported in Section 3.

Table 3. Three selected synthetic bindings illustrating the result format of **Listing 1**; this is not a complete query result.

Student resource	Academic record	Academic year	Department
student-001	record-001	academic-year-2023	department-001
student-002	record-002	academic-year-2024	department-002
student-003	record-003	academic-year-2025	department-003

3.5 Evaluation Measures

For each workload and worker configuration, execution time was measured from SPARQL request submission until the complete response was returned. The primary measure was end-to-end response time in seconds. Result-set row counts were also recorded to verify that equivalent requests were compared across repetitions.

For each configuration, the arithmetic mean, median, sample standard deviation, minimum, maximum, interquartile range, and normal-approximation 95% confidence interval were calculated from the measured repetitions. The confidence interval is reported as the mean plus or minus its half-width (CI_{hw}), calculated using **Eq. 1**.

$$CI_{hw} = 1.96 \times \frac{s}{\sqrt{n}} \qquad \text{(Eq. 1)}$$

where s is the sample standard deviation and n is the number of measured repetitions ($n = 40$). The mean estimates typical end-to-end response time.

The evaluation compares 1 to 7 workers descriptively. No formal hypothesis test was performed; therefore, differences between configurations are reported as observed differences rather than statistically significant improvements. All successful measurements from the primary benchmark were retained, and no post-hoc outlier deletion was applied. The use of repeated measurements and descriptive variability statistics follows established guidance for computer-system performance evaluation [11].

4. RESULTS AND DISCUSSION

All 840 requests in the primary benchmark completed successfully. Deterministic query ordering produced the expected and stable result sizes across repetitions: 9 rows for the SMIS-only workload and 20 rows for each federated workload. **Table 4** summarises end-to-end response time using the median and interquartile range for each worker configuration. In this notation, 0.421 [0.387, 0.457] s means that the median response time was 0.421 s, while the middle 50% of measured response times fell between the first quartile (0.387 s) and the third quartile (0.457 s). **Fig. 4** shows the full response-time distributions.

The 7-worker SMIS-Portal configuration includes one successful 22.478-second request. This measurement was retained, and no post-hoc outlier deletion was applied.

4.1 Query Performance Across Worker Configurations

The three workloads had distinct response time profiles. The lowest observed mean response time for the SMIS-only workload occurred with one worker at 0.427 ± 0.014 s. Its observed mean increased to between 0.488 s and 0.585 s for 2 to 6 workers, showing no benefit from additional distributed execution for this small single-source workload.

Table 4. End-to-end response times (s) across 40 successful repetitions per worker configuration.[5]

Workers	SMIS-Only	SMIS-Portal	SMIS-RMIS/RIC
1	0.421 [0.387, 0.457] / 0.427 ± 0.014	1.166 [1.131, 1.234] / 1.214 ± 0.064	0.527 [0.499, 0.555] / 0.529 ± 0.011
2	0.470 [0.434, 0.531] / 0.488 ± 0.020	1.147 [1.080, 1.273] / 1.166 ± 0.047	0.661 [0.622, 0.716] / 0.670 ± 0.025
3	0.493 [0.440, 0.550] / 0.510 ± 0.027	1.081 [1.017, 1.214] / 1.128 ± 0.053	0.687 [0.627, 0.763] / 0.707 ± 0.032
4	0.545 [0.505, 0.613] / 0.556 ± 0.024	1.121 [1.031, 1.208] / 1.143 ± 0.059	0.729 [0.674, 0.802] / 0.754 ± 0.036
5	0.569 [0.520, 0.600] / 0.583 ± 0.028	1.115 [1.054, 1.208] / 1.144 ± 0.038	0.786 [0.724, 0.884] / 0.825 ± 0.046
6	0.590 [0.521, 0.628] / 0.585 ± 0.024	1.151 [1.051, 1.204] / 1.140 ± 0.042	0.819 [0.761, 0.883] / 0.842 ± 0.032
7	0.541 [0.500, 0.576] / 0.546 ± 0.023	1.118 [1.017, 1.203] / 1.634 ± 1.048	0.841 [0.803, 0.879] / 0.839 ± 0.027

[5] Each cell reports median [Q1, Q3] / mean ± 95% confidence interval.

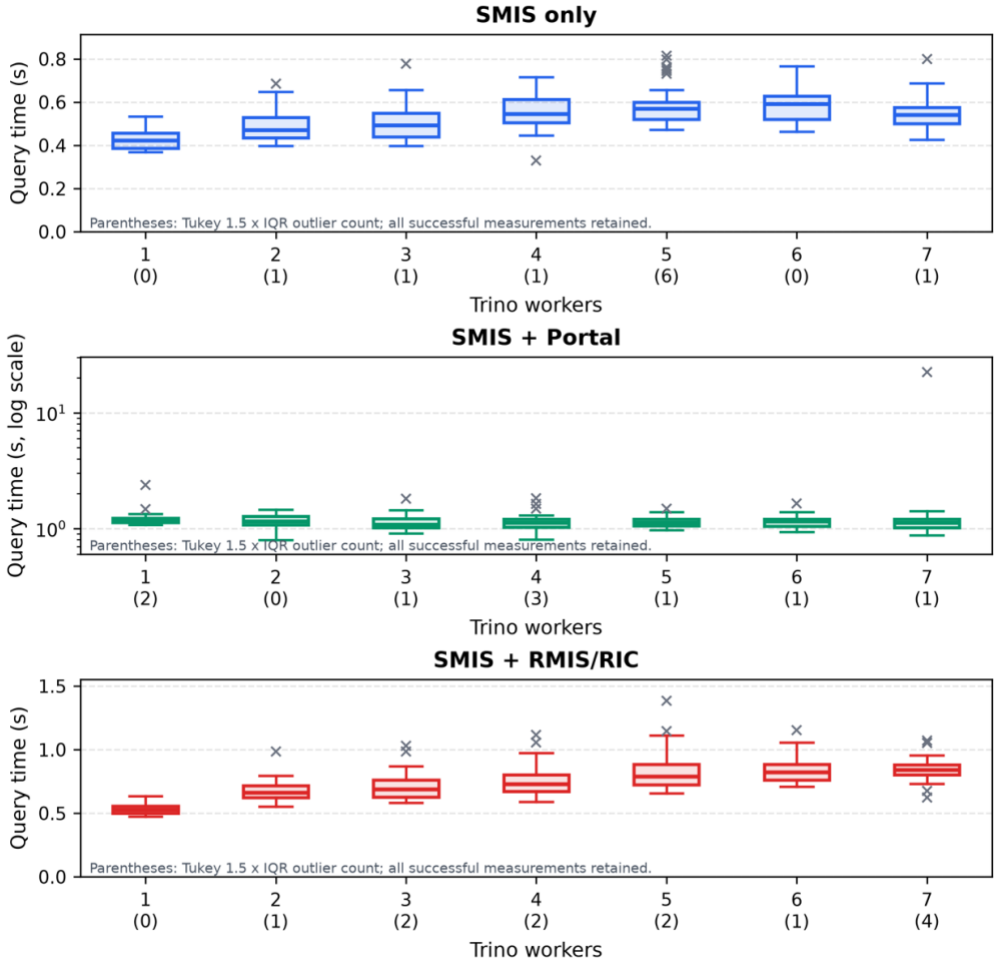


Fig. 4. Response-time distributions under the direct Ubuntu Server VM deployment.[6]

The SMIS-Portal workload was the slowest workload. Its lowest observed mean was 1.128 ± 0.053 s with three workers, compared with 1.214 ± 0.064 s with one worker. The results for one to six workers were similar, while the 7-worker mean was affected by one retained long-running request. Four follow-up repetitions of the 7-worker configuration were conducted, each with 40 measurements per workload. The 22.478-second SMIS-Portal observation did not recur; the maximum SMIS-Portal response times were 1.441 s, 1.594 s, 1.372 s, and 1.274 s. One separate 4.757-second SMIS-RMIS/RIC request occurred in a follow-up repetition. Container inspection after the final repetition reported no OOM kills, container restarts, or non-zero exit codes. These observations indicate intermittent tail latency rather than sufficient evidence to remove the original successful measurement from the primary analysis.

The SMIS-RMIS/RIC workload was fastest with one worker at 0.529 ± 0.011 s. The observed mean increased as worker count increased, reaching 0.839 ± 0.027 s with 7 workers.

These comparisons are descriptive. No formal hypothesis test was performed; therefore, the observed differences must not be interpreted as statistically significant performance improvements.

4.2 Practical Scalability and System Bottlenecks

The results show that the effect of adding Trino workers depends on workload characteristics and can be negative for small workloads. The SMIS-only query returns a small result set from one source, so the fixed costs of query planning, SPARQL rewriting, and service communication are likely

[6] Box plots show medians, interquartile ranges, and Tukey outliers; all successful measurements are retained. The SMIS-Portal panel uses a logarithmic y-axis to display both typical response times and the retained long-running observation.

more important than parallel execution. Adding workers cannot substantially reduce these fixed costs.

The federated workloads require access to multiple institutional sources and cross-source query processing. The SMIS-Portal workload was slower than the other workloads. Possible explanations include the combined effect of Ontop query rewriting, Trino planning and scheduling, PostgreSQL processing, cross-source joins, and result transfer. The measurements should therefore be interpreted as end-to-end performance of the deployed VKG rather than performance of Trino or Ontop in isolation.

Potential optimisation strategies include reviewing source-table indexes used by join conditions, simplifying expensive mappings, caching frequently requested results, and selectively materialising high-demand data.

4.3 Discussion of Findings

The findings support the feasibility of the proposed architecture for the evaluated use case. The Trino- and Ontop-based VKG provided direct, non-materialised semantic access to distributed ITC research-information systems without copying the operational databases into a separate RDF store. All primary benchmark requests completed successfully and returned stable result sizes through the same SPARQL endpoint.

The architecture is suitable for read-only administrative integration and exploratory semantic queries where response times in the measured range are acceptable. It prioritises direct and non-materialised access to the operational sources and reduced data replication. However, it should not be treated as a replacement for a locally materialised RDF store when an application requires guaranteed sub-second response times or high concurrent query throughput.

4.4 Limitations and Threats to Validity

The evaluation used three representative read-only workloads, forty repetitions per configuration, and one deployment environment. The results may not generalise to larger result sets, more complex graph patterns, high query concurrency, or different network conditions. The direct server deployment provided direct source connectivity; however, the study does not isolate the effect of network path because the server hardware and deployment configuration differ from earlier VPN-based measurements reported in [17]. Future evaluation may complement the institution-specific workloads with standardised SPARQL benchmark workloads, such as the Berlin SPARQL Benchmark [1].

The study compares worker configurations descriptively. It does not provide a formal statistical hypothesis test, and it does not measure query throughput under concurrent users. Future experiments should include controlled network-latency measurements, concurrent workloads, larger result sets, source-level execution plans, and repeated benchmarking over longer periods.

5. CONCLUSIONS

This study implemented and evaluated a Virtual Knowledge Graph (VKG) for federating research-information systems at the Institute of Technology of Cambodia. The architecture combines institutional PostgreSQL sources, Trino as a federated SQL layer, Ontop as an Ontology-Based Data Access engine, and a SPARQL endpoint. It provides a shared semantic view of researchers, academic records, projects, and research-related information while retaining the original operational databases and avoiding a separate materialised RDF store.

The evaluation considered three representative workloads: a single-source SMIS workload, a federated SMIS-Portal workload, and a federated SMIS-RMIS/RIC workload. All 840 requests in the primary benchmark completed successfully and returned stable result sizes. The results confirm that the proposed VKG can provide practical direct and non-materialised semantic access to distributed institutional data through a common SPARQL interface.

The worker-scaling evaluation did not show a consistent reduction in response time as the number of Trino workers increased from 1 to 7. One worker produced the lowest observed mean response time for the SMIS-only and SMIS-RMIS/RIC workloads, while three workers produced the lowest observed mean for the SMIS-Portal workload. No formal hypothesis test was performed. The results therefore do not establish a statistically significant scaling benefit for the evaluated low-volume workloads. The SMIS-Portal workload remained slower than the other workloads. This result indicates that workload-specific joins, source access, and distributed-query processing remain important end-to-end considerations.

The proposed architecture is appropriate for read-only administrative integration and exploratory semantic queries where direct, non-materialised access and reduced data replication are more important than guaranteed sub-second response times for the evaluated workloads and deployment. Future work should evaluate higher query concurrency, larger result sets, more complex SPARQL patterns, source-level indexing, caching, and selective materialisation of frequently requested information. It should also investigate alignment with established research-information models [3,12] and external identity matching, which requires a validated reference alignment for reliable evaluation [9]. More broadly, the architecture follows the objective of data integration: providing a unified access layer across autonomous heterogeneous sources [13].

DATA AND CODE AVAILABILITY

Sanitised ontology files, Ontop mappings, deterministic SPARQL benchmark queries, benchmark scripts, and aggregated evaluation results are available at: https://github.com/Phearun2020/itc-rmis-kg. Institutional source data are not publicly available because they contain confidential administrative and research information.

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